



KOLEJ MATRIKULASI JOHOR

LAB TASK REPORT DATABASE 2 (CD125) SEMESTER 2 2025/2026

LECTURER'S NAME:

LATIFAH BINTI ZAKARIA

TOPIC:

APPLICATION OF DATA IN COMPUTING PROJECT

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S2

LAB TASK

Topic 4 : Application of Data in Computing Project

4.2 SDLC for project development

(12 hours)

Theme: Smart Attendance System

Individual attendance records are important information for an organization as proof of an individual's participation or presence in every activity organized. Manual attendance recording often causes several difficulties such as inaccurate information, scattered records, overlapping data, unclear details, and others.

As a system developer, you are required to develop a web-based application that can record or verify individual attendance in every activity organized by the organization. The system developed must be accessible using a web browser from any computer within the Local Area Network (LAN).

LAB TASK REPORT

PLANNING (Initial assessment & feasibility study)

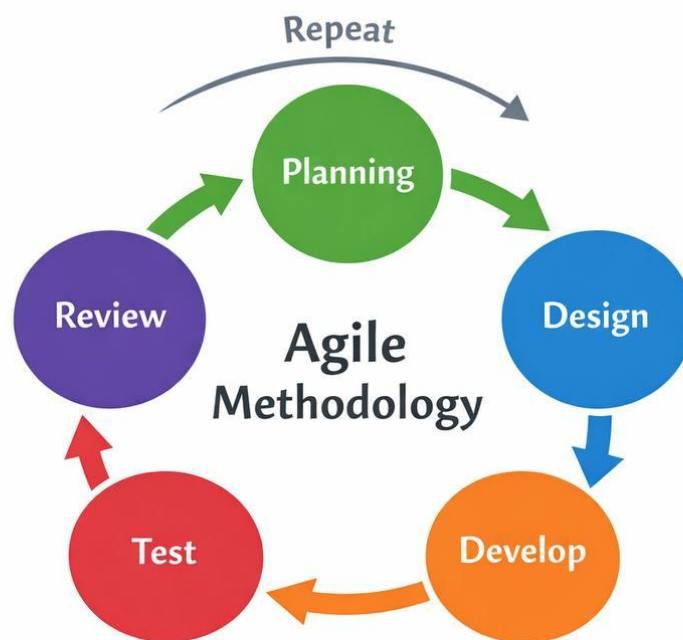
4.2 (i) Select appropriate SDLC methodology for computing project development

Agile methodology is chosen for this project because it supports iterative and incremental system development through multiple sprints. Instead of developing the system in a single sequence, the system is built step by step, where each sprint focuses on specific modules such as login, attendance management, and reporting. This methodology allows:

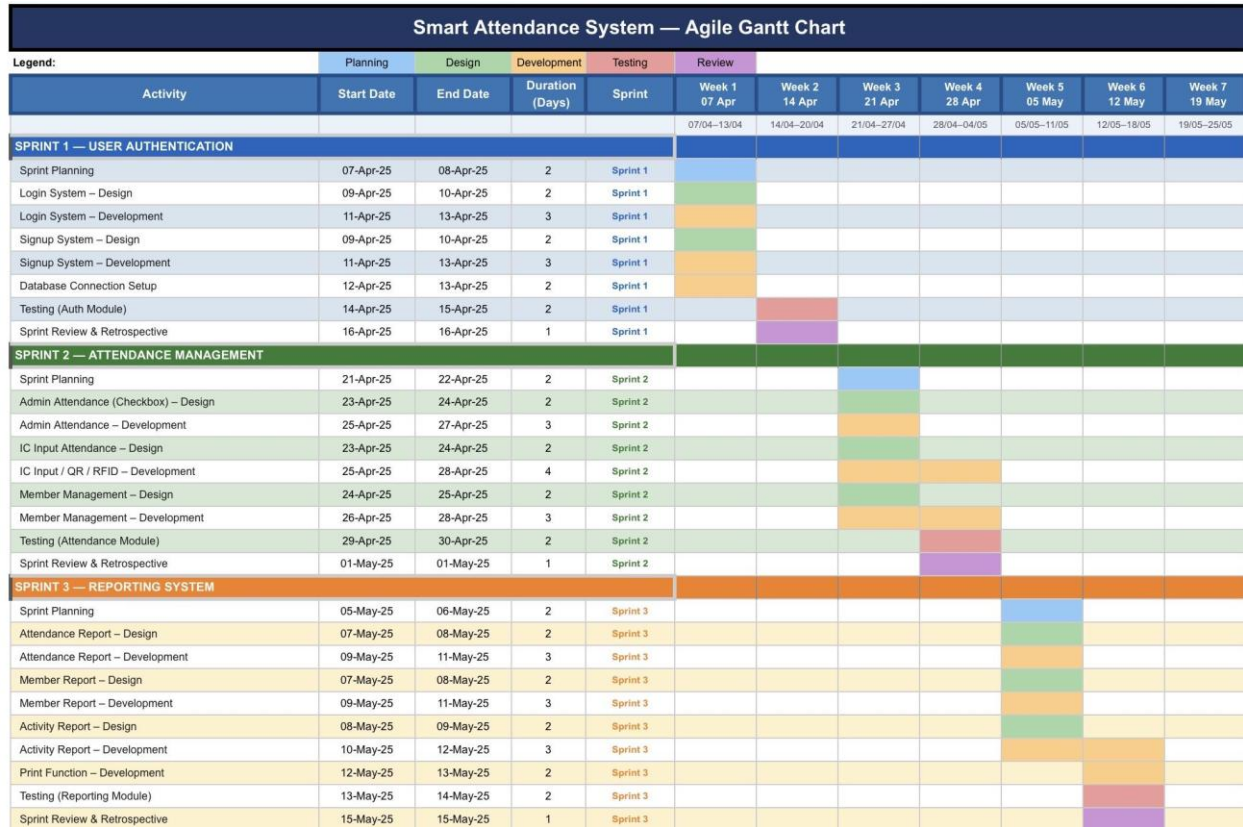
- Continuous testing and debugging in every iteration
- Immediate feedback from users after each sprint
- Progressive improvement of system features

In this project, Agile is implemented using three sprints, where each sprint follows the cycle: Planning → Design → Develop → Test → Review

Agile methodology allows continuous improvement through iterative feedback from users in each sprint.



4.2 (ii) Develop plan (GANTT CHART)



ANALYSIS (User requirements, existing system evaluation & logical system design)	
4.2 (iii) Identify problem	
1.1 Identify Problem	In addition, there are several problems such as inaccurate attendance information, where the total attendance may not be calculated correctly due to manual counting, increasing the risk of human error. Furthermore, data duplication may occur because the records are written manually, leading to the same data being recorded more than once and causing inconsistency. Lastly, the recorded information may be unclear due to illegible handwriting from members, making it difficult for teachers or club secretaries to accurately identify and verify attendance.
1.2 Objectives	The objectives of this system are: 1. Record the attendance of SMK Tinggi Klang Ping Pong Club members at each meeting. 2. Display analysis and attendance list for each activity meeting. 3. Record the attendance of each without missing records as the system ensures all attendance is properly recorded and managed.
1.3 Scope	The scopes of this system are: 1. This system is only capable of recording the attendance of members of the Ping Pong Club at SMK Tinggi Klang. 2. This system will only display attendance records and attendance based on each meeting. 3. The system is able to record attendance using RFID / QR code / barcode. The system only allows the Ping Pong Club teacher to record attendance.
1.4 Target Group	This system is only used by the administrator, who is the teacher advisor of the Ping Pong Club at SMK Tinggi Klang, and the users, who are the members of the Ping Pong Club at SMK Tinggi Klang only.

4.2 (iv) Analyze requirements

1.5 Evaluating Existing Systems

Since the current attendance system is done manually and does not involve any cost, attendance is recorded using a pen and attendance record book.

Weakness :

- This system takes a long time for teachers or club secretaries to analyse attendance records and may lead to errors in calculation.

Strength:

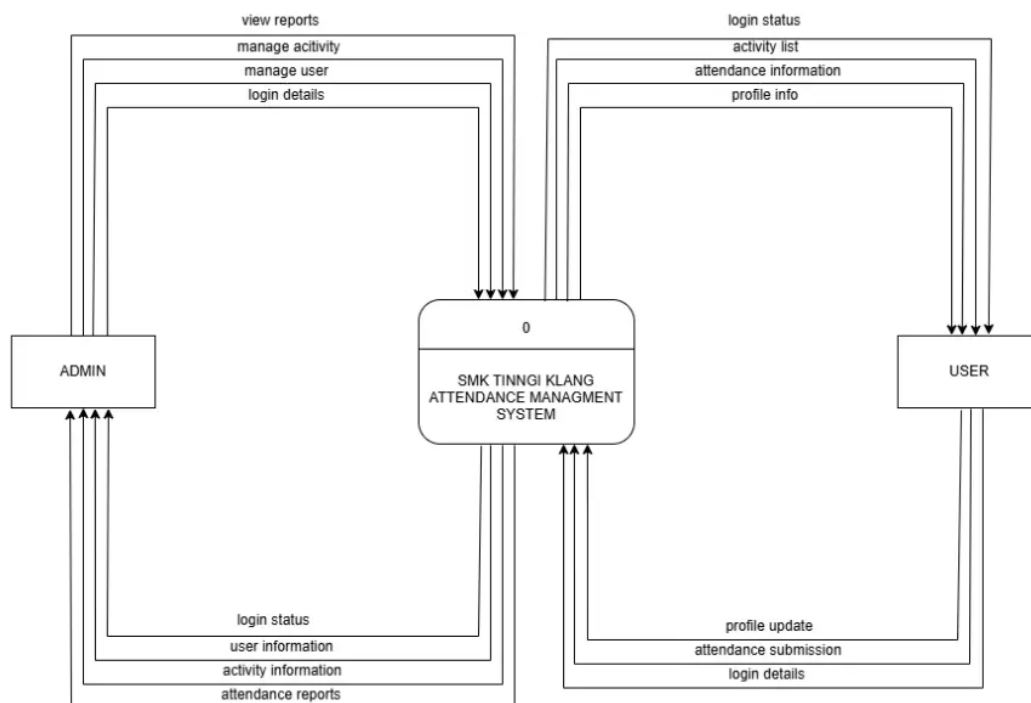
- This system does not require internet access to record attendance.
- This system does not involve high costs as it only uses a pen and an attendance record book.

New system browser justification:

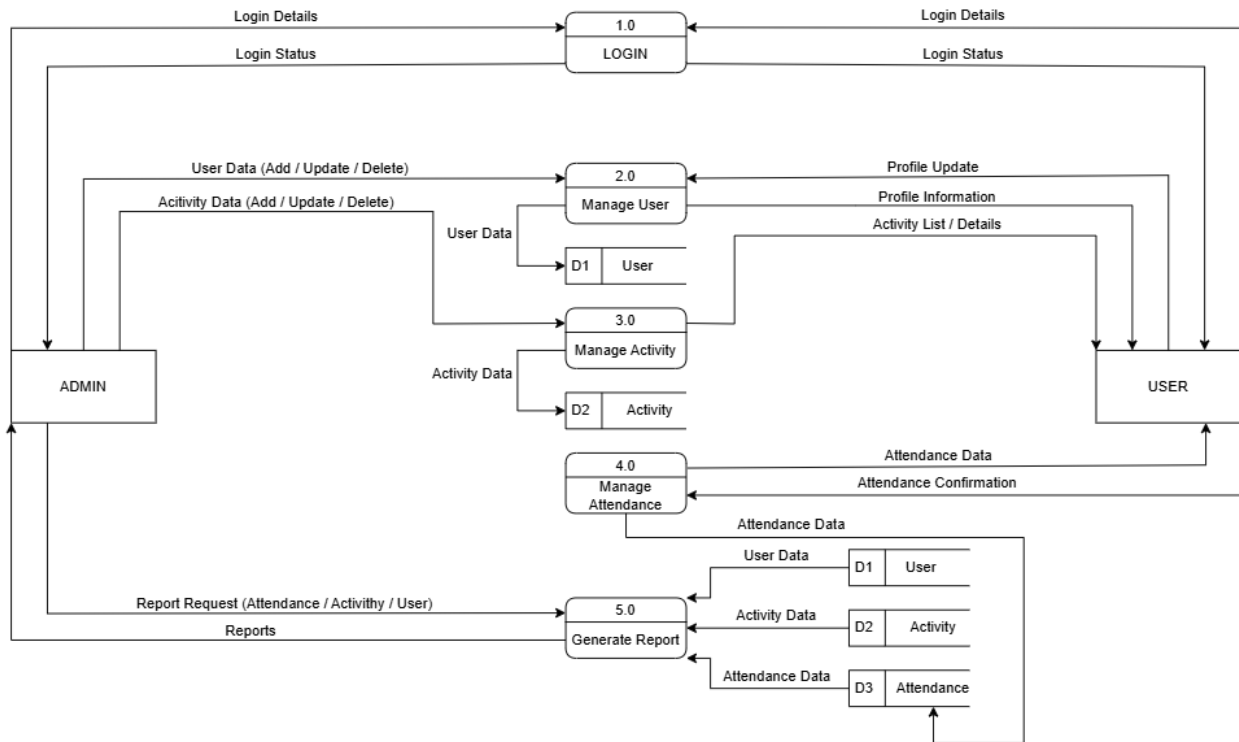
Therefore, a new system needs to be developed to help teachers obtain attendance analysis quickly and accurately, as well as to speed up the process of recording and verifying club members' attendance.

System name: Ping Pong Club Attendance System SMK Tinggi Klang

DFD (CONTEXT DIAGRAM)



DFD (LEVEL 0)

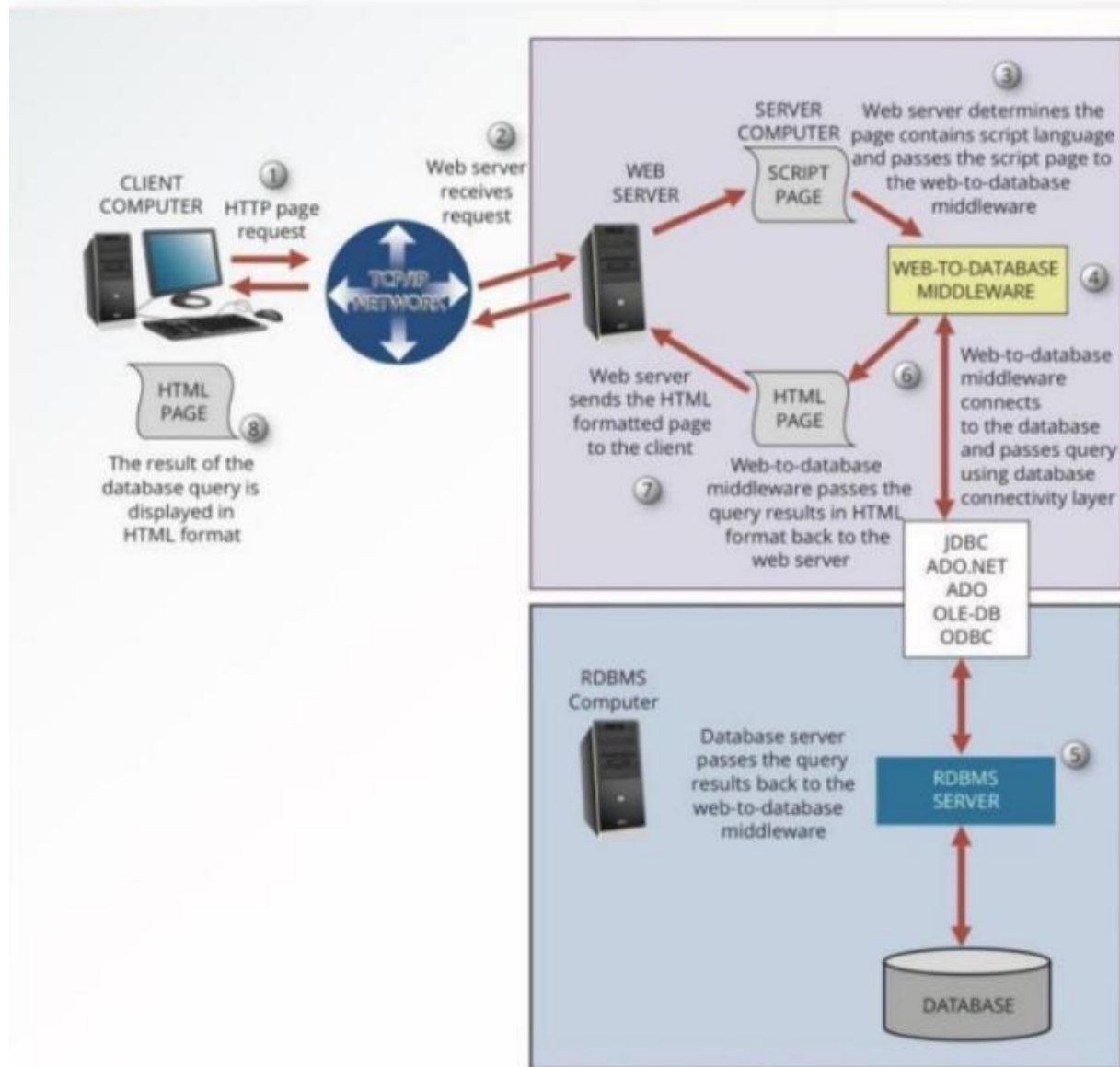


DETAILED SYSTEMS DESIGN (Detailed system specification)

4.2 (v) Design

ARCHITECTURE

Figure 15.8 Web-to-Database Middleware

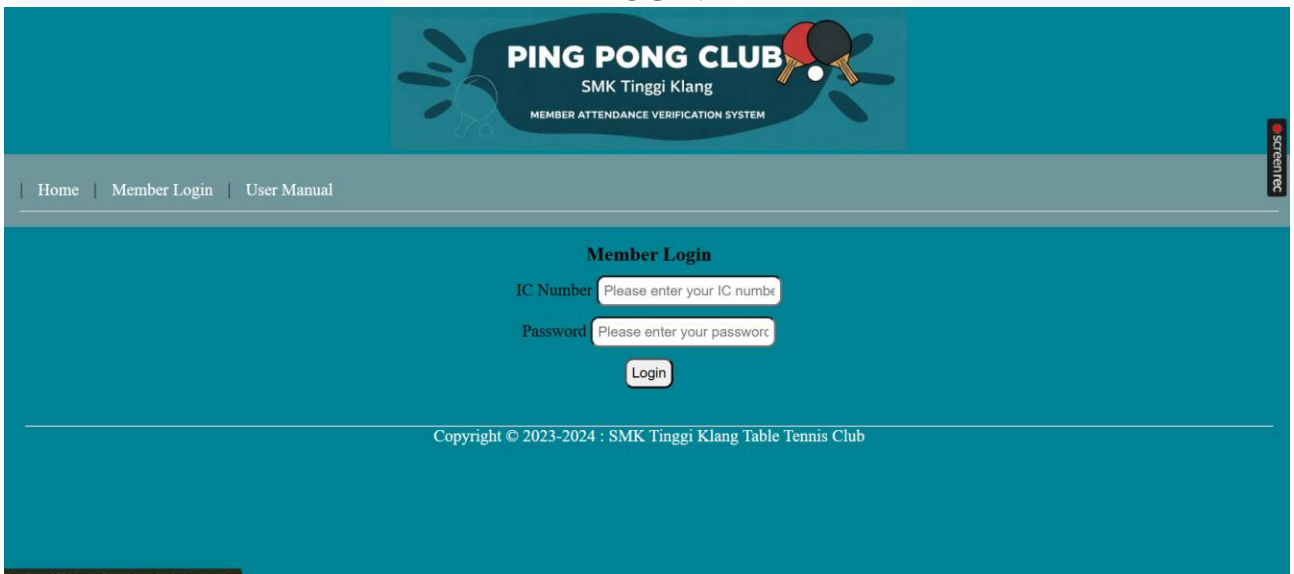


USER INTERFACE

MENU

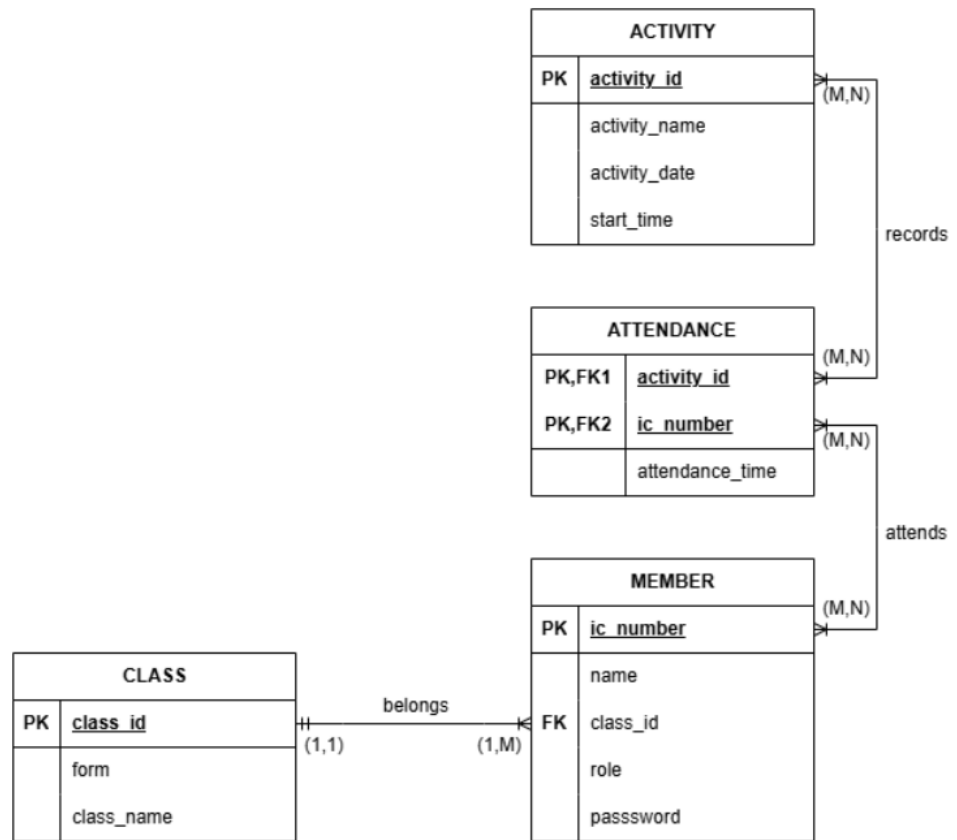


LOGIN



DATABASE (conceptual/logical)

(i) Conceptual
***use crows foot



(ii) Logical
(scheme)

ATTENDANCE (activity_id<PK><FK>, ic_number<PK><FK>, attendance_time)
MEMBER (ic_number<PK>, name, class_id <FK>, role, password)
ACTIVITY (activity_id<PK>, activity_name, activity_date, start_time)
CLASS (class_id<PK>, form, class_name)

(iii) Data Dictionary

***copy or
screenshot from
XAMPP for all
tables**

attendance_system

activity

Column	Type	Null	Default	Links to	Comments	Media type
activity_id (Primary)	int(2)	No				
activity_name	text	Yes	NULL			
activity_date	date	Yes	NULL			
start_time	varchar(20)	Yes	NULL			

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	activity_id	0	A	No	

attendance

Column	Type	Null	Default	Links to	Comments	Media type
activity_id (Primary)	int(2)	No		activity -> activity_id		
ic_number (Primary)	varchar(12)	No		member -> ic_number		
attendance_time	time	Yes	NULL			

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	activity_id	0	A	No	
				ic_number	0	A	No	
ic_number	BTREE	No	No	ic_number	0	A	No	

class

Column	Type	Null	Default	Links to	Comments	Media type
class_id (Primary)	int(2)	No				
form	varchar(2)	Yes	NULL			
class_name	varchar(60)	Yes	NULL			

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	class_id	4	A	No	

member

Column	Type	Null	Default	Links to	Comments	Media type
name	varchar(60)	Yes	NULL			
ic_number (Primary)	varchar(12)	No				
class_id	int(2)	Yes	NULL	class -> class_id		

Column	Type	Null	Default	Links to	Comments	Media type
role	varchar(20)	Yes	NULL			
password	varchar(15)	Yes	NULL			

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	ic_number	5	A	No	
class_id	BTREE	No	No	class_id	5	A	Yes	

AGILE ITERATION (SPRINT SECTION)
(Iterative development, testing and continuous improvement)

4.2 (vi) Agile Development (Sprint Iterations)

Sprint 1: User Authentication (Login & Signup)

This sprint focuses on developing the core access control of the system, ensuring that only authorised users can access system functionalities.

Objective:

The primary objective of Sprint 1 is to design and implement a secure user authentication module that controls access to the system. This includes the development of login and registration (signup) functionalities, along with session-based access control to differentiate between Admin and Student roles.

Planning:

During the planning phase, the system requirements for authentication were identified. The system supports two types of users:

- Admin (teacher or club advisor)
- Student (club member)

Each user is required to log in using a unique 12-digit IC number and password.

The database structure was planned with:

- member table (ic_number, name, class_id, role, password)
- class table (class_id, form, class_name)

A central database connection file (connection.php) was also planned. In addition, access control files (kawalan-admin.php and kawalan-biasa.php) were designed to restrict page access based on user roles.

Design:

The user interface was designed to be simple and user-friendly.

- index.php acts as the landing page with navigation links
- login-borang.php provides input fields for IC number and password
- signup-borang.php includes fields for name, IC number, class, role, and password

The navigation bar dynamically changes based on user session:

- Guests → Home, Login, User Manual
- Admin → Full system access
- Student → Limited access (Profile only)

Development:

The authentication module was developed using several PHP files:

- connection.php

Establishes connection to the MySQL database using `mysqli_connect()` and ensures system can access stored data.

- login-proses.php

Processes login data using POST method. It validates IC number and password against the database. If valid, session variables are created and the user is redirected. Invalid input triggers an error message.

- signup-proses.php

Handles user registration. The system validates:

- IC number must be 12 digits and numeric
- IC number must not already exist

If validation passes, user data is inserted into the database.

- logout.php

Destroys the session using `session_destroy()` and redirects the user to the homepage.

- kawalan-admin.php / kawalan-biasa.php

Restrict access to pages based on user roles by checking session variables.

Testing:

Testing was conducted to ensure all authentication functions operate correctly.

- Valid login credentials successfully grant system access
- Invalid IC number or password displays error message
- Empty input fields are rejected
- Duplicate IC number during signup is prevented
- Role-based navigation is verified (Admin vs Student)
- Logout function successfully destroys session and prevents access to restricted pages

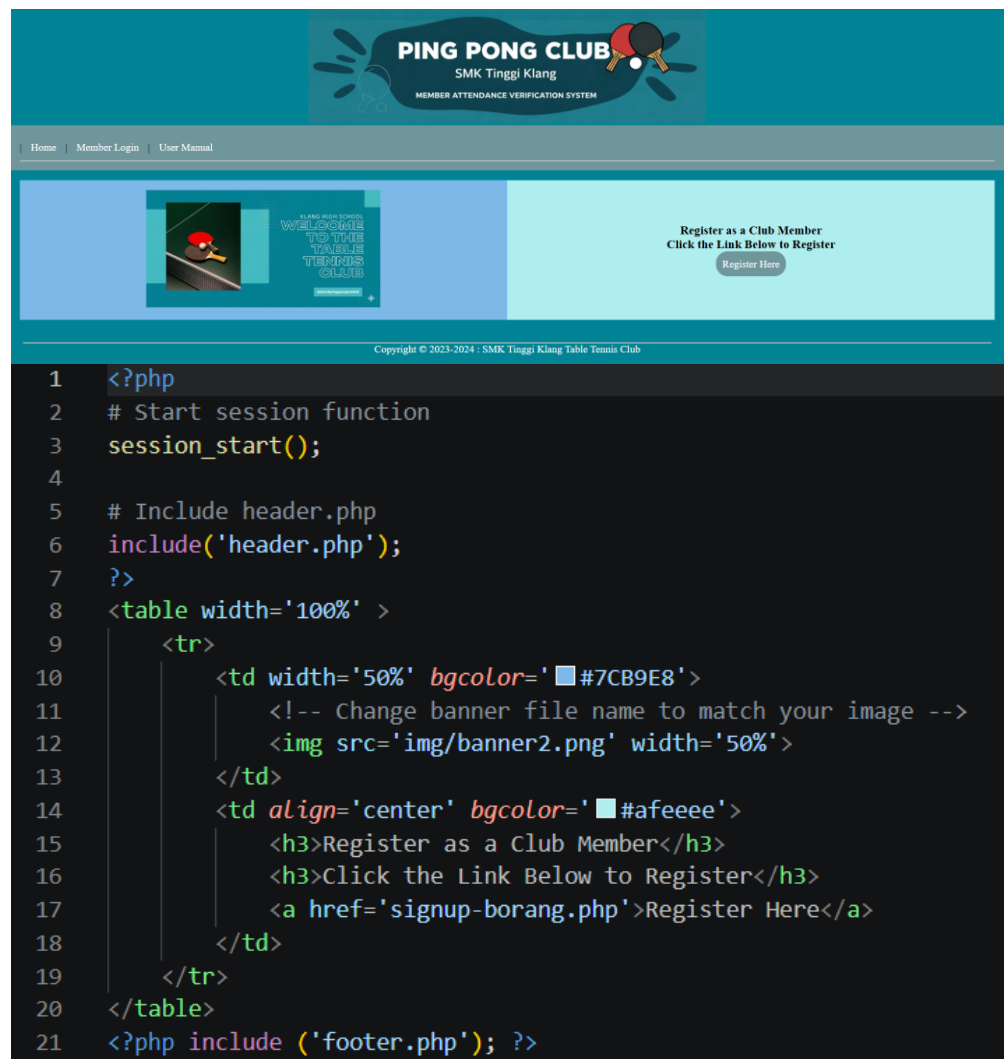
All test cases confirmed that no invalid data was stored in the database.

Review:

The Sprint 1 review confirmed that all authentication features were successfully implemented. The system is able to securely manage user registration, login validation, session handling, and role-based access control.

This sprint establishes the foundation for subsequent modules, allowing iterative improvements in future sprints in accordance with Agile methodology.

index.php (Home / Landing Page):



This page is the main entry point of the system. It displays the club information and navigation menu for users. Unauthenticated users can only access limited options such as login and registration.

login-borang.php (Login Page):



```

1 <?php
2 # Start session function
3 session_start();
4
5 # Include header.php
6 include('header.php');
7 ?>
8
9 <!-- Login page title -->
10 <h3>Member Login</h3>
11
12 <!-- Login form -->
13 <form action='login-proses.php' method='POST'>
14
15     IC Number    <input type='text'      name='ic_number' placeholder='Please enter your IC number'><br>
16     Password     <input type='password' name='password' placeholder='Please enter your password'><br>
17                 <input type='submit'   value='Login'>
18 </form>
19 <?php include ('footer.php'); ?>

```

This interface allows users to log into the system using their IC number and password.

The system validates the input with the database to ensure secure access.

signup-borang.php (Registration Page):

```

1 <?php
2 # Start SESSION function
3 session_start();
4
5 # Include header.php & connection.php
6 include('header.php');
7 include('connection.php');
8 ?>
9
10 <!-- Page title -->
11 <h3> New Member Registration </h3>
12
13 <!-- New Member Registration Form -->
14 <form action = 'signup-proses.php' method = 'POST'>
15
16     Member Name    <input type ='text'      name ='name' placeholder='Please enter your name'      required> <br>
17     IC Number      <input type ='text'      name ='ic_number' placeholder='Please enter your IC number' required> <br>
18     Class           <select name='class_id'><br>
19                     <option selected disabled value>Please Select Class</option>
20                     <?php
21                         # Display list of classes in dropdown
22                         $sql_select_class = "select* from class";
23                         $execute_select_class = mysqli_query($condb,$sql_select_class);
24                         while($m=mysqli_fetch_array($execute_select_class))
25                         {
26                             echo "<option value='". $m['class_id']."'>
27                                 ".$m['form']. " ". $m['class_name']. "
28                             </option>";
29                         }
30                     ?>
31                     </select> <br>
32     Role            <select name='role' required><br>
33                     <option selected disabled value>Please Select Role</option>
34                     <option value='Admin'>Admin</option>
35                     <option value='Student'>Student</option>
36                     </select> <br>

```

This interface allows new users to register by entering their personal details. The class dropdown is dynamically retrieved from the database. The system prevents duplicate IC numbers.

Admin Navigation Bar (After Login):



This interface shows the navigation menu available for Admin users after login. Admin has full access to system features such as attendance management and reports.

Student Profile Page:




```

1 <?php
2 # Start session function
3 session_start();
4
5 # Include header.php and connection.php
6 include('header.php');
7 include('connection.php');
8
9 # Check if session 'ic_number' exists
10 if(empty($_SESSION['ic_number'])){
11
12     # If session ic_number is empty, stop execution
13     die("<script>alert('Please login');
14         window.location.href='logout.php';</script>");
15 }
16 ?>
17
18 <table width='100%' bgcolor='#afeeee' border='1'>
19     <tr>
20         <td width='70%' align='center' valign='top' >
21             <h3 >Attendance Record</h3>
22
23             <!-- Table header for activity list -->
24             <table align='center' width='100%' border='1' id='saiz' bgcolor='white'>
25                 <caption>
26                     Self-confirmation can only be done on the date the activity takes place
27                 </caption>
28                 <tr align='center' bgcolor='#00BED7' >
29                     <td>Activity Name</td>
30                     <td>Date | Time</td>
31                     <td>Attendance</td>
32                 </tr>
33             </table>
34         </td>
35         <td>
36             # Query to fetch activity list
37             $sql_select_activities="select* from activity";

```

This interface shows the student's personal profile and attendance record. Students have limited access and cannot access administrative features.

Sprint 2: Attendance Management

This sprint focuses on developing the core functionality of the system, which is attendance recording and management.

Objective:

The objective of this sprint is to implement a complete attendance management module that allows attendance to be recorded accurately and efficiently using multiple methods.

Planning:

During the planning phase, two attendance recording methods were identified:

- Admin-controlled attendance (bulk confirmation)

Admin selects members who attended an activity using checkboxes.

- Real-time attendance (IC number input)

Attendance is recorded by entering or scanning the member's IC number.

The system also includes:

- Activity management (create, update, delete)
- Member management (update, delete, bulk upload)

The database design ensures:

- Each attendance record is unique using a composite key (activity_id + ic_number)
- Attendance is linked to both member and activity tables

- Attendance time is recorded automatically

Design:

Several interfaces were designed for attendance management:

- Activity List Page (senarai-aktiviti.php)

Displays all activities with options to update, delete, or record attendance

- Attendance Confirmation Form (kehadiran-borang.php)

Shows list of members with checkboxes for Admin to mark attendance

- Attendance Counter Page (kehadiran-rekod.php)

Allows IC number input for real-time attendance recording

- Member List Page (senarai-ahli.php)

Displays all members with management functions

- Student Profile Page (profil.php)

Shows individual attendance record and allows self-confirmation on activity day

Development:

The attendance module was implemented using multiple processes:

- Bulk Attendance (Admin Confirmation)

The system uses a delete-then-insert approach:

- Existing attendance records for an activity are deleted
- New records are inserted based on selected checkboxes
- Attendance time is recorded using date() function
- Real-time IC Recording (Attendance Counter)

The system performs validation before inserting data:

- Check if IC exists in member table
- Check for duplicate attendance for same activity
- Insert attendance record if valid

Status messages are displayed:

- Green → Attendance confirmed
- Red → Invalid IC or duplicate entry
- Member Management

Admin can update, delete, and upload members in bulk

- Student Self Confirmation

Students can confirm their own attendance on the activity date via profile page.

Testing:

Testing was conducted to ensure all attendance features function correctly:

- Bulk attendance correctly saves selected members
- Existing attendance records are pre-selected
- Duplicate attendance is prevented
- Invalid IC numbers are rejected
- Real-time attendance updates correctly in the system

- Activity CRUD operations function correctly
- Student self-confirmation works only on valid dates

All scenarios were tested to ensure data accuracy and system reliability.

Review:

The Sprint 2 review confirmed that the attendance management module was successfully implemented.

Both attendance methods (bulk admin confirmation and real-time IC input) function correctly. The system ensures data accuracy by preventing duplicate entries and validating all inputs.

Feedback from this sprint is used to improve system usability and accuracy in the next iteration, following Agile principles.

senarai-aktiviti.php (Activity List Page):



Activity List		
Activity Name	Date Time	Action
Mesyuarat Agung Kelab Ping Pong	2024-01-13 08:00 AM	Update Delete Attendance Confirmation
Sesi Latihan Asas Ping Pong	2024-01-27 02:00 PM	Update Delete Attendance Confirmation
Latihan Kemahiran Servis & Smash	2024-02-10 02:00 PM	Update Delete Attendance Confirmation
Pertandingan Dalam Kelab (Pusingan 1)	2024-02-24 09:00 AM	Update Delete Attendance Confirmation

```

1 <?php
2 # Start session function
3 session_start();
4
5 # Include header.php, connection.php and kawalan-admin.php
6 include('header.php');
7 include('connection.php');
8 include('kawalan-admin.php');
9
10 ?>
11 <h3 align='center'>Activity List</h3>
12
13 <!-- Table header for activity list -->
14 <table align='center' width='100%' border='1' id='saiz'>
15     <tr bgcolor='#00B8D2'>
16         <td>
17             <form action='' method='POST' style="margin:0; padding:0;">
18                 <input type='text' name='activity_name' placeholder='Search Activity'>
19                 <input type='submit' value='Search'>
20             </form>
21         </td>
22         <td colspan='2' align='right'>
23             | <a href='aktiviti-daftar-borang.php'>Register New Activity / Meeting</a> |
24             <?php include('butang-saiz.php'); ?>
25         </td>
26     </tr>
27     <tr bgcolor='#00BED7' align='center'>
28         <td>Activity Name</td>
29         <td>Date | Time</td>
30         <td>Action</td>
31     </tr>
32
33 <?php
34
35 # Additional condition for activity search query
36 $extra="";

```

This page displays all activities in a table format. Admin can update, delete, or select an activity to record attendance. A search function is also provided.

kehadiran-borang.php (Attendance Confirmation Form):



[Home](#) | [Profile](#) | [Attendance Counter](#) | [Member List](#) | [Activity List](#) | [Attendance Report](#) | [Member Report](#) | [Activity Report](#) | [Logout](#)

Member Attendance Confirmation
 Activity Name : Meysuan Agung Kelab Ping Pong
 Date | Time : 2024-01-13 | 08:00 AM

Change Text Size

No.	Name	IC Number	Class	Attendance
1	Aminul Hakim	070304100195	5 AK	☒
2	ERNA	070913100654	4 SH	☒
3	Farhana Husna	060819100674	4 SH	☒

```

1 <?php
2 # Start session function
3 session_start();
4
5 # Include header.php, connection.php and kawalan-admin.php
6 include('header.php');
7 include('connection.php');
8 include('kawalan-admin.php');
9
10 # Fetch activity data from database
11 $sql_activity = "select * from activity where activity_id
12 ='".$_GET['activity_id']."' ";
13 $execute_activity = mysqli_query($condb,$sql_activity);
14 $n = mysqli_fetch_array($execute_activity);
15
16 ?>
17
18 <h3>Member Attendance Confirmation</h3>
19
20 Activity Name : <?= $n['activity_name'] ?> <br>
21 Date | Time : <?= $n['activity_date']." | ".$n['start_time'] ?><br>
22 <br><br>
23
24 <?php include('butang-saiz.php'); ?>
25 <form action='kehadiran-proses.php?activity_id=<?= $_GET['activity_id'] ?>'
26 method='POST'>
27 <table border='1' id='saiz' width='100%'>
28 <tr>
29 <td>No.</td>
30 <td>Name</td>
31 <td>IC Number</td>
32 <td>Class</td>
33 <td>Attendance</td>
34 </tr>
35
36 <?php

```

This interface allows Admin to record attendance by selecting members using checkboxes. Previously recorded attendance is automatically pre-selected.

kehadiran-rekod.php (Attendance Counter Page):

```

1 <?php
2 # Start session function
3 session_start();
4
5 # Include external files and declare variables
6 include('header.php');
7 include('kawalan-admin.php');
8 include('connection.php');
9 $time=date("H:i:s");
10 $status=""; # Used to display attendance status
11 $color=""; # Used for status background color
12
13 # Check if POST ic_number exists
14 if(!empty($_POST['ic_number'])){
15     # Check if the entered IC number exists in the database
16     $sql_check = "select* from member where ic number = '".$_POST['ic_number']."' ";
17     $execute_check = mysqli_query($condb,$sql_check);
18     if(mysqli_num_rows($execute_check)!=1)
19     {
20         # IC number not found in system
21         $status ="The IC number entered/scanned is not in the system";
22         $color="red";
23     } else {
24         # Check if the IC number has already recorded attendance
25         $sql_check_att = "select* from attendance where ic_number='".$_POST['ic_number']."'
26         and activity_id = '".$_GET['activity_id']."' limit 1";
27
28         $execute_check_att = mysqli_query($condb,$sql_check_att);
29         if(mysqli_num_rows($execute_check_att)==1)
30         {
31             $status ="You have already confirmed your attendance previously.";
32             $color="red";
33         } else {
34             # Save attendance record
35             $save_data=mysqli_query($condb,"insert into attendance
36

```

This interface allows attendance to be recorded by entering or scanning the member's IC number. The system validates the IC and records attendance in real time.

senarai-ahli.php (Member List Page):

[Home](#) | [Profile](#) | [Attendance Counter](#) | [Member List](#) | [Activity List](#) | [Attendance Report](#) | [Member Report](#) | [Activity Report](#) | [Logout](#)

Member List

| Change Text Size:

Name	IC Number	Class	Password	Role	Action
Farhana Husna	060819100674	4 SH	123	Student	Update Delete
ERNA	070913100654	4 SH	123	Admin	Update Delete
Syafiq	071124100563	4 SH	123	Student	Update Delete
Hafizuddin	070512100321	4 SK	123	Student	Update Delete

```

1  <?php
2  # Start session function
3  session_start();
4
5  # Include header.php, connection.php and kawalan-admin.php
6  include('header.php');
7  include('connection.php');
8  include('kawalan-admin.php');
9
10 ?>
11 <!-- Header for member list table -->
12 <h3 align='center'>Member List</h3>
13
14 <table align='center'width='100%'border='1'id='saiz'>
15     <tr bgcolor='#00B8D2'>
16         <td colspan='3'>
17             <form action='' method='POST' style="margin:0; padding:0;">
18                 <input type='text' name='name' placeholder='Search Member Name'>
19                 <input type='submit' value='Search'>
20             </form>
21         </td>
22         <td colspan='3' align='right'>
23             | <a href='upload.php'>Upload Members</a> |
24             <?php include('butang-saiz.php'); ?>
25         </td>
26     </tr>
27     <tr bgcolor='#00BED7'>
28         <td width='35%'>Name</td>
29         <td width='15%'>IC Number</td>
30         <td width='10%'>Class</td>
31         <td width='10%'>Password</td>
32         <td width='10%'>Role</td>
33         <td width='20%'>Action</td>
34     </tr>
35
36 <?php

```

This page allows Admin to manage member data, including updating, deleting, and uploading member records.

profil.php (Student Profile Page):

PING PONG CLUB
SMK Tinggi Klang
MEMBER ATTENDANCE VERIFICATION SYSTEM

Home | Profile | Logout

Attendance Record
Self-confirmation can only be done on the date the activity takes place

Activity Name	Date Time	Attendance
Measyuarat Agung Kelab Ping Pong	2024-01-13 08:00 AM	✓
Sesi Latihan Asas Ping Pong	2024-01-27 02:00 PM	✓
Latihan Kemahiran Servis & Smash	2024-02-10 02:00 PM	✓
Pertandingan Dalaman Kelab (Pusingan 1)	2024-02-24 09:00 AM	✓
Gotong Royong Bilik Kelab	2024-03-02 08:00 AM	✓
Latihan Intensif Pra-Pertandingan	2024-03-16 02:00 PM	✓

SCAN CODE TO CONFIRM ATTENDANCE
Name : Nansha
IC Number : 060703100757

This interface displays the student's attendance record. Students can also confirm their attendance on the activity date.

Sprint 3: Reporting System

This sprint focuses on developing reporting functionalities that allow administrators to analyse and generate attendance data efficiently.

Objective:

The objective of this sprint is to develop a reporting system that provides clear and structured attendance information, enabling administrators to monitor participation and generate reports.

Planning:

During the planning phase, three types of reports were identified:

- Activity Attendance Report

Displays attendance status for each member in a selected activity

- Member Report

Displays a list of all registered members

- Activity Report

Displays summary of attendance for all activities

Each report includes:

- Search/filter functionality
- Print function for documentation

The system retrieves data from multiple tables:

- member
- activity
- attendance
- class

Design:

All report pages were designed with a consistent layout:

- A navigation bar for system access

- A search bar for filtering data
- A structured table format for displaying data
- A “Print Report” button

Special design considerations:

- Attendance status is shown using symbols (✓ / ✗)
- Summary section displays totals and percentages
- Print layout uses CSS to hide unnecessary elements

Development:

The reporting module was developed using database queries and dynamic data display:

- Activity Attendance Report (kehadiran-laporan.php)
- Retrieves attendance data using JOIN between tables
- Displays attendance status (present/absent)
- Calculates total attendance and percentage
- Member Report (laporan_ahli.php)
- Displays all members with class and role
- Supports search by member name
- Activity Report (laporan_aktiviti.php)
- Displays all activities with attendance count
- Calculates attendance percentage for each activity

Additional features:

- Search function implemented using SQL LIKE
- Data validation using mysqli_real_escape_string()
- Print function using window.print()

Testing:

Testing was conducted to ensure all reports function correctly:

- Attendance data displays accurately
- Attendance percentage calculations are correct
- Search function filters data correctly
- Reports display correct number of records
- Print function produces clean output
- Symbols (✓ / ✗) display correctly

All reports were tested under different scenarios to ensure accuracy and reliability.

Review:

The Sprint 3 review confirmed that all reporting features were successfully implemented.

The system is able to generate accurate and structured reports for attendance,

members, and activities. The search and print functionalities enhance usability and support documentation needs.

This final sprint integrates all previous modules and ensures the system meets user requirements through continuous improvement in line with Agile methodology.

kehadiran-laporan.php (Activity Attendance Report):



The screenshot displays the 'Activity Attendance Report' page of the 'PING PONG CLUB SMK Tinggi Klang' system. The header includes a navigation menu with links: Home, Profile, Attendance Counter, Member List, Activity List, Attendance Report, Member Report, Activity Report, and Logout. The main content area shows a dropdown menu for 'Activity' with the option 'Please Select Activity', a 'Search' button, and the selected activity 'Mesyuarat Agung Kelab Ping Pong'. Below this, the date '2024-01-13 | 08:00 AM' and attendance statistics 'Attendance : 9 / 10' and 'Percentage : 90.00 %' are displayed. A table with 5 columns (No., Name, IC Number, Class, Attendance) lists three members: Amirul Hakim (5 | AK), ERNA (4 | SH), and Farhana Husna (4 | SH), all with green checkmarks in the Attendance column. The table has a yellow header and a light blue body. To the right of the table, there are controls for 'Change Text Size' (reset, +, -) and a 'Print' button.

No.	Name	IC Number	Class	Attendance
1	Amirul Hakim	070304100195	5 AK	✓
2	ERNA	070913100654	4 SH	✓
3	Farhana Husna	060819100674	4 SH	✓

```

1  <?php
2  # Start session function
3  session_start();
4
5  # Include header.php, connection.php and kawalan-admin.php
6  include('header.php');
7  include('connection.php');
8  include('kawalan-admin.php');
9
10 ?>
11 <h3>Activity Attendance Report</h3>
12 <!-- Activity search form -->
13 <form action='' method='GET'>
14     Activity <select name='activity_id'>
15         <option selected disabled value>Please Select Activity</option>
16
17         <?php
18         # Display activity list in dropdown
19         $sql_activities = "select* from activity";
20         $execute_activities = mysqli_query($condb,$sql_activities);
21         while($n = mysqli_fetch_array($execute_activities))
22         {
23             echo "<option value='". $n['activity_id']."'>
24                 ". $n['activity_id']. " | ". $n['activity_name']. "
25                 </option>";
26         }
27         ?>
28     </select>
29     <input type='submit' value='Search'>
30 </form>
31
32 <?php
33 $extra="";
34 if(!empty($_GET['activity_id']))
35 {
36     # Get activity_id from URL

```

This page displays attendance records for a selected activity. It shows each member's attendance status along with total attendance and percentage.

kehadiran-laporan.php (Search Function):

The screenshot displays the 'Activity Attendance Report' page of the PING PONG CLUB SMK Tinggi Klang. The page features a navigation bar with links: Home, Profile, Attendance Counter, Member List, Activity List, Attendance Report, Member Report, Activity Report, and Logout. The main content area has a search bar for 'Search Member Name' and a dropdown menu for 'Activity'. The dropdown menu lists 12 activities, including 'Mesyuarat Agung Kelab Ping Pong', 'Sesi Latihan Asas Ping Pong', 'Latihan Kemahiran Servis & Smash', 'Pertandingan Dalamann Kelab (Pusingan 1)', 'Gotong Royong Bilik Kelab', 'Latihan Intensif Pra-Pertandingan', 'Pertandingan Antara Kelas Ping Pong', 'Lawatan Ke Pusat Ping Pong Negeri', 'Bongkai Teknik Permainan Bersama Jurulatih', 'Pertandingan Persahabatan Antara Sekolah', 'Latihan Penutup & Penilaian Ahli', and 'Majlis Penutup & Penyampaian Hadiah Kelab'. Below the search bar, there is a table with columns: No., Name, IC N, and Attendance. The table shows three members: Amirul Hakim (IC 070304), ERNA (IC 070913), and Farhana Husna (IC 060819100674). The attendance for each member is marked with a green checkmark.

No.	Name	IC N	Attendance
1	Amirul Hakim	070304	✓
2	ERNA	070913	✓
3	Farhana Husna	060819100674	✓

This interface allows the user to search for specific members within the attendance report using a keyword.

laporan_ahli.php (Member Report):

The screenshot displays the 'Member Report' page of the PING PONG CLUB SMK Tinggi Klang. The page features a navigation bar with links: Home, Profile, Attendance Counter, Member List, Activity List, Attendance Report, Member Report, Activity Report, and Logout. The main content area has a search bar for 'Search Member Name' and a 'Print Report' button. Below the search bar, there is a table with columns: No., Name, IC Number, Class, and Role. The table shows six members: ERNA (IC 070913100654, Class 4 | SH, Admin), Farhana Husna (IC 060819100674, Class 4 | SH, Student), Syafiq (IC 071124100563, Class 4 | SH, Student), Hafizuddin (IC 070512100321, Class 4 | SK, Student), Syukri (IC 070826100813, Class 4 | SK, Student), and Zulhimi (IC 071005100887, Class 4 | SK, Student).

No.	Name	IC Number	Class	Role
1	ERNA	070913100654	4 SH	Admin
2	Farhana Husna	060819100674	4 SH	Student
3	Syafiq	071124100563	4 SH	Student
4	Hafizuddin	070512100321	4 SK	Student
5	Syukri	070826100813	4 SK	Student
6	Zulhimi	071005100887	4 SK	Student

```

1 k?php
2 # Start session function
3 session_start();
4
5 # Include header.php, connection.php and kawalan-admin.php
6 include('header.php');
7 include('connection.php');
8 include('kawalan-admin.php');
9
10 ?>
11
12 <style>
13 @media print {
14     nav, .no-print { display: none !important; }
15     body { background-color: white !important; }
16     header { background-color: white !important; color: black !important; }
17 }
18 </style>
19
20 <h3>Member Report</h3>
21
22 <!-- Member search form -->
23 <table align='center' width='100%' border='1'>
24 <tr bgcolor='#0088D2'>
25 <td class='no-print'>
26 <form action='' method='POST' style='margin:0; padding:0;'>
27 <input type='text' name='name' placeholder='Search Member Name'
28 value='<?= !empty($_POST['name']) ? htmlspecialchars($_POST['name']) : "" ?>'>
29 &nbsp;<input type='submit' value='Search'>
30 </form>
31 </td>
32 <td align='right' class='no-print'>
33 <a href='javascript:window.print()' style='background-color:#008394;'>&#128424; Print Report</a>
34 </td>
35 </tr>
36 <tr bgcolor='yellow'>

```

This page displays a list of all registered members along with their class and role. A search function is provided for filtering data.

laporan_aktiviti.php (Activity Report):

 PING PONG CLUB SMK Tinggi Klang MEMBER ATTENDANCE VERIFICATION SYSTEM							
Home Profile Attendance Counter Member List Activity List Attendance Report Member Report Activity Report Logout							
Activity Report							
Search Activity Name Search		Print Report					
No.	Activity Name	Date	Time	Total Attendance	Total Members	Attendance Rate	
1	Majlis Penutup & Penyampaian Hadiah Kelab	2024-05-18	09:00 AM	10	10	100.00 %	
2	Latihan Penutup & Penilaian Ahli	2024-05-04	02:00 PM	8	10	80.00 %	
3	Pertandingan Persahabatan Antara Sekolah	2024-04-27	08:00 AM	7	10	70.00 %	
4	Bengkel Teknik Permainan Bersama Jurualatih	2024-04-13	09:00 AM	8	10	80.00 %	
5	Lawatan Ke Pusat Ping Pong Negeri	2024-04-06	07:30 AM	9	10	90.00 %	
6	Pertandingan Antara Kelas Ping Pong	2024-03-23	08:30 AM	8	10	80.00 %	

```

1 <?php
2 # Start session function
3 session_start();
4
5 # Include header.php, connection.php and kawalan-admin.php
6 include('header.php');
7 include('connection.php');
8 include('kawalan-admin.php');
9
10 ?>
11
12 <style>
13 @media print {
14     nav, .no-print { display: none !important; }
15     body { background-color: white !important; }
16     header { background-color: white !important; color: black !important; }
17 }
18 </style>
19
20 <h3>Activity Report</h3>
21
22 <!-- Activity search form -->
23 <table align='center' width='100%' border='1'>
24 <tr bgcolor='#008B8D'>
25 <td class='no-print'>
26 <form action='' method='POST' style='margin:0; padding:0;'>
27 <input type='text' name='activity_name' placeholder='Search Activity Name'
28 value='<?> empty($_POST['activity_name']) ? htmlspecialchars($_POST['activity_name']) : "" ?>'>
29 &nbsp;  <input type='submit' value='Search'>
30 </form>
31 </td>
32 <td align='right' class='no-print'>
33 <a href='javascript:window.print()' style='background-color:#008394;'>&#128424; Print Report</a>
34 </td>
35 </tr>
36 <tr bgcolor='yellow'>

```

This page displays all activities with their attendance statistics, including total attendance and percentage.

Print Preview (Report Output):

05/04/2026, 17:26
ATTENDANCE SYSTEM - SMK TINGGI KLANG TABLE TENNIS CLUB



Activity Report

No.	Activity Name	Date	Time	Total Attendance	Total Members	Attendance Rate
1	Majlis Penutup & Penyampaian Hadiah Kelab	2024-05-18	09:00 AM	10	10	100.00 %
2	Latihan Penutup & Penilaian Ahli	2024-05-04	02:00 PM	8	10	80.00 %
3	Pertandingan Persahabatan Antara Sekolah	2024-04-27	08:00 AM	7	10	70.00 %
4	Bengkel Teknik Permainan Bersama Jurualatih	2024-04-13	09:00 AM	8	10	80.00 %
5	Lawatan Ke Pusat Ping Pong Negeri	2024-04-06	07:30 AM	9	10	90.00 %
6	Pertandingan Antara Kelas Ping Pong	2024-03-23	08:30 AM	8	10	80.00 %
7	Latihan Intensif Pra-Pertandingan	2024-03-16	02:00 PM	7	10	70.00 %
8	Gotong Royong Bilik Kelab	2024-03-02	08:00 AM	10	10	100.00 %
9	Pertandingan Dalamn Kelab (Pusingan 1)	2024-02-24	09:00 AM	9	10	90.00 %
10	Latihan Kemahiran Servis & Smash	2024-02-10	02:00 PM	10	10	100.00 %

localhost\attendance\han\app\open_aktiviti.php
1/2

Print
2 sheets of paper

Destination
Microsoft Print to PDF

Pages
All

Layout
Portrait

Colour
Colour

More settings

Print
Cancel

```

1  <!-- Function to change text size for accessibility -->
2  <script>
3
4  function changeSize(multiplier) {
5      // Get current font size of the table
6      var sizeEl = document.getElementById("saiz");
7      var current_size = sizeEl.style.fontSize || 1;
8
9      /* Check which button was pressed:
10         reset button - value 2 sent - return to original size
11         +   button - value 1 sent - increase text size
12         -   button - value -1 sent - decrease text size
13     */
14     if(multiplier==2)
15     {
16         sizeEl.style.fontSize = "1em";
17     }
18     else
19     {
20         sizeEl.style.fontSize = ( parseFloat(current_size) + (multiplier * 0.2)) + "em";
21     }
22 }
23 </script>
24
25 <!-- Buttons to change text size -->
26 | Change Text Size |
27 <input name='reSize1' type='button' value='reset'          onclick="changeSize(2)" />
28 <input name='reSize'  type='button' value='&nbsp;+&nbsp;'         onclick="changeSize(1)" />
29 <input name='reSize2' type='button' value='&nbsp;-&nbsp;'         onclick="changeSize(-1)" />
30 |
31 <button onclick="window.print()">Print</button>

```

This interface shows the print preview of the report. The system hides unnecessary elements to produce a clean and structured output.

Testing			
(i) Syntax Error		No Syntax error	
(ii) Logic Error			
Testing was conducted across all modules including login, attendance recording, and reporting to ensure system accuracy and reliability.			
Item's Name	Data's Name	Data sample	Output (Screenshot)
Data Type	VARCHAR	070913109285	New Member Registration Member Name WONG IC Number 070913109285 Class 4 SH Role Admin Password ... Register localhost says Registration Successful. Please Login OK
Lower Limit	12	07091310	localhost says Error: Invalid IC Number OK
Upper Limit	12	0108284832722	localhost says Error: Invalid IC Number OK
Run Time Error			
No run time error OR State in the section where there is an error.			

MAINTENANCE

(Evaluation, maintenance and enhancement)

In the maintenance phase, the system needs to go through an evaluation process to identify weaknesses and then make enhancements to improve the system so that it is more effective and meets user needs.

Evaluation : After the system was deployed, an evaluation was conducted by observing how Admin and Student users interact with the system. Feedback was gathered from the club teacher and members regarding the usability of features such as member registration, attendance recording, and report generation. During evaluation, several weaknesses were identified — the registration form had no email field for communication, the attendance report could not be exported to PDF or Excel, and the system had no password encryption which posed a security risk.

Maintenance : Based on the evaluation findings, maintenance was carried out to fix the identified issues. The database was updated to use hashed passwords (MD5/bcrypt) instead of storing plain text passwords in the member table. The IC number validation in `signup-proses.php` was also tightened to prevent duplicate entries. Minor bug fixes were applied to the attendance counter page where the status color did not reset properly between sessions.

Enhancement : Several enhancements were made to improve the system's effectiveness and meet user needs. An email field was added to the member registration form to allow the Admin to contact members directly. A PDF/Excel export button was added to the Attendance Report page so the teacher advisor can print records easily. Additionally, a dashboard was introduced on the Home page for Admin to view a quick summary of total members, total activities, and overall attendance percentage — making the system more informative and efficient for club management.